Draft

1 A Model of Chokepoint Bargaining

The model has three players: Europe (E), the United States (U), and China (C). Europe controls whether ASML's tools, spare parts, and service engineers reach Chinese fabs. In practice that control is split among the Dutch licensing authority, the firm, and the European Union, and we take the split up in Section 3.1. Until then we treat Europe as a unitary actor. The model is built to answer one question: what determines the price Europe can charge the United States for keeping ASML on the Atlantic side of the AI race?

1.1 Setup

Alignment. Europe chooses an alignment $a \in \{A, X\}$. Under Atlantic alignment (A), ASML supplies only US-aligned fabs and complies with every US denial requirement. Under exit (X), Europe lets ASML supply China with tools, servicing, or licences that the United States wants withheld. Exit stands for any realignment that moves frontier lithography toward China. Section 3.3 adds an intermediate option, hedging, which corresponds to the legacy immersion DUV sales and servicing at issue in the 2023–2025 Dutch export controls and in the scenario's 2028 crisis.

Stakes. The United States loses $V > 0$ if Europe exits. V measures the patron's dependence on the chokepoint: what America gives up in the AI race if China gains access to ASML's tools. China gains $W > 0$ from Europe's exit. A share $\theta \in [0, 1]$ of that gain can be blocked by US jurisdiction over China-bound flows, through the foreign direct product rule, de minimis rules for tools containing US-origin content, and control of US-made subsystems. Enforcing that jurisdiction against China-bound shipments costs the United States little, so we treat it as credible throughout and let θ capture its reach.

Promises. Each great power can offer Europe a transfer conditional on alignment. The US offer $\tau_U \geq 0$ is paid if Europe chooses A; China's offer $\tau_C \geq 0$ is paid if Europe chooses X. Transfers are in kind: tier status under an AI export regime, exemptions from extraterritorial rules, tariff relief, market access, credit, or co-production. Promises must be self-enforcing, so we cap them at B_U and B_C, the largest transfers each power would still deliver after Europe has moved. Europe values a unit of China's in-kind offer at $\lambda_C \in (0, 1]$, which captures both the inefficiency of in-kind transfers and doubts about delivery. The most China can credibly put on the table, in units of European utility, is therefore

$$\omega \equiv \lambda_C \min\{(1 - \theta)W,\, B_C\}. \tag{1}$$

Linked instruments. After Europe chooses its alignment, each great power decides how to use the instruments it controls. These decisions are moves in the game, so a threat matters only if carrying it out is sequentially rational. There are four instruments.

Security. With some probability a security crisis occurs in which Europe needs US protection. The United States defends Europe if its stake in doing so, s_a, exceeds a crisis-specific cost K drawn from a continuous distribution G. Europe loses $S > 0$ if it is left undefended. The probability of defence after alignment a is $\pi_a = G(s_a)$. We allow $s_A \geq s_X$: a Europe that supplies China's chipmakers may be worth less to Washington. The *conditionality* of the guarantee is

$$\kappa \equiv \pi_A - \pi_X = G(s_A) - G(s_X) \in [0, 1]. \tag{2}$$

Frontier AI. The United States can deny Europe access to frontier models and inference, which costs Europe $\alpha \geq 0$. Its ex post gain from denial after alignment a is $g_{ai}(a) = \nu - k_a$. Here ν is the opportunity value of the compute released, which is high when compute is scarce, and k_a is the value to the United States of continuing to supply Europe given its alignment, with $k_A > k_X \geq 0$. If the United States has publicly committed to deny frontier AI to states that realign, failing to apply that rule to an exiting Europe costs it $m_{ai} \geq 0$ in audience costs (**?**); supplying a loyal Europe violates no commitment.

Firm-level penalties. The United States can penalize ASML itself, by entity listing, denial of US-origin inputs, or withdrawal of US-person support, which costs Europe $f \geq 0$. Because ASML also supplies America's own fabs and those of its allies, penalizing the firm harms the United States. We write its ex post gain from penalizing an exiting Europe as $g_f = \gamma - \varphi V$, where γ is the added denial benefit and φV the self-inflicted cost. The associated audience cost is $m_f \geq 0$.

Chinese coercion. China can restrict exports of critical inputs, such as rare earths, magnets, or the legacy chips at issue in the 2025 Nexperia dispute, which costs Europe $r \geq 0$. Punishment reverses nothing in a one-shot game, so China carries it out only if its ex post value, g_r, which includes its reputation with other chokepoint holders, plus any audience cost m_r is nonnegative.

Values. Europe bears an intrinsic cost $\eta \geq 0$ from exit, reflecting the value it places on keeping frontier AI capabilities away from an authoritarian rival that supports Russia's war.

Timing. (1) The United States offers $\tau_U \in [0, B_U]$. (2) China observes τ_U and offers $\tau_C \in [0, B_C]$. (3) Europe chooses a. (4) Transfers are paid, nature determines whether a crisis occurs and draws K, and each power chooses how to use its instruments. Offers lie on a grid of mesh $\varepsilon > 0$ and we report limits as $\varepsilon \to 0$; Europe breaks ties in favour of A, and China does not bid when it would earn zero. The United States moves first because, in the scenario and in the 2023–2024 record, Washington sets the terms and others respond (cf. **?**). Section 1.4 shows which results depend on this choice.

Payoffs. We normalize payoffs relative to Atlantic alignment with no transfers. The United States receives $-\tau_U$ if Europe chooses A and $-V$ if Europe exits, where V is net of whatever the United States gains from its ex post responses. China receives $(1 - \theta)W - \tau_C$ if Europe exits and zero otherwise. Europe's payoffs are stated in eq. (5) below, once we know which instruments are used.

1.2 Credible conditionality

Only the part of each instrument that depends on Europe's alignment can enter Europe's choice. The first step is to find which threats survive sequential rationality. Define the indicators

$$\chi_{ai} \equiv \mathbb{1}\{\nu + m_{ai} \geq k_X\}\mathbb{1}\{\nu < k_A\}, \qquad \chi_f \equiv \mathbb{1}\{\varphi V \leq \gamma + m_f\}, \qquad \chi_r \equiv \mathbb{1}\{g_r + m_r \geq 0\}. \tag{3}$$

Lemma 1 (Credible conditionality)**.** *In every subgame-perfect equilibrium, after alignment a: (i) the United States defends Europe in a crisis if and only if $s_a \geq K$; (ii) it denies frontier AI after exit if and only if $\nu + m_{ai} \geq k_X$, and after Atlantic alignment if and only if $\nu \geq k_A$; (iii) after exit it penalizes ASML if and only if $\varphi V \leq \gamma + m_f$, and it never penalizes ASML after A; (iv) China punishes Europe if and only if $a = A$ and $g_r + m_r \geq 0$. Relative to Atlantic alignment, exit therefore costs Europe*

$$c \equiv \kappa S + \chi_{ai}\,\alpha + \chi_f f + \eta, \tag{4}$$

and Atlantic alignment costs Europe $\chi_r r$ in Chinese punishment.

Part (ii) shows that the AI instrument is conditional only inside a scarcity window. When compute is plentiful ($\nu + m_{ai} < k_X$), the United States sells to Europe whatever its alignment. When compute is extremely scarce ($\nu \geq k_A$), it rations even a loyal Europe, as the scenario's 2029 Frontier Inference Services Rule does to most of the EU. In both cases denial does not depend on Europe's choice and drops out of c. Part (iii) is where the patron's dependence on the chokepoint enters: the more the United States needs ASML, the less credible a threat to cripple it. Jurisdiction over China-bound flows, captured by θ, has no such limit.

Substituting Lemma 1, Europe's payoffs are

$$u_E(A) = \tau_U - \chi_r r, \qquad u_E(X) = \lambda_C \tau_C - c, \tag{5}$$

up to terms common to both alignments, which affect Europe's welfare but not its choice.

Definition 1 (Exit premium)**.** Europe's *exit premium* is the most Europe could gain by exiting, net of the credible costs of exit, relative to Atlantic alignment with no US transfer:

$$Q \equiv \omega - c + \chi_r r = \lambda_C \min\{(1-\theta)W, B_C\} - \kappa S - \chi_{ai}\alpha - \chi_f f - \eta + \chi_r r. \tag{6}$$

The term follows **?**: a state's influence over a partner depends on what the partner would lose by leaving. In our setting the relevant quantity runs the other way, from the chokepoint holder's side. Q prices Europe's credible alternative, and the US dependence V appears in it only through χ_f.

1.3 Equilibrium: the exit premium sets the price

Proposition 1 (Leverage equals the exit premium)**.** *Suppose $Q \leq \min\{V, B_U\}$. As $\varepsilon \to 0$, every subgame-perfect equilibrium has Europe choosing A, with the US transfer converging to*

$$L \equiv [Q]^+ = \max\{0, Q\}. \tag{7}$$

Europe's payoff converges to $[Q]^+ - \chi_r r$, the United States' to $-[Q]^+$, and China's to zero. If instead $Q > \min\{V, B_U\}$, Europe exits.

We call L Europe's *leverage*: the transfer it extracts for keeping the chokepoint aligned. The logic is that of an auction in which the losing bidder sets the price. China can induce exit only by offering Europe more than the United States does, net of exit costs, and it will spend at most what exit is worth to it. The United States therefore pays exactly enough to make China's best credible offer unattractive. When $Q \leq 0$, even China's best offer cannot outweigh the costs the United States can credibly impose and the value Europe places on its alignment, so the United States pays nothing.

Three features of Proposition 1 drive the rest of the paper. First, V does not appear in L except through χ_f. A chokepoint the patron cannot do without is worth no more to its holder than one the

patron barely needs, as long as the holder's credible alternative is the same. Second, the threshold $Q > 0$ is a credibility condition. It asks whether Europe would actually prefer China's best credible package to Atlantic alignment with nothing, once US responses are taken into account. Third, the condition $Q \leq B_U$ matters. If the United States cannot credibly promise enough, Europe exits even when keeping it aligned is worth more to Washington than exit is worth to Beijing.

1.4 Which results depend on who moves first

A bargaining theorist will ask whether V drops out only because the United States proposes first. It partly does. Proposition 2 collects the cases.

Proposition 2 (Protocol bounds)**.** *Assume ASML keeps supplying US-aligned fabs while bargaining continues (no hold-up), $V \geq Q$, and $B_U \geq V$. Let $L_0 \equiv [\chi_r r - c]^+$ and $\bar{L} \equiv \min\{V, [\lambda_C B_C - c + \chi_r r]^+\}$, the largest transfer a Chinese offer within its cap could force. Europe's leverage L is:*

(a) US offers first, China second: $L = [Q]^+$ (Proposition 1).

(b) Simultaneous offers: the Nash equilibrium set is $L \in [[Q]^+, \bar{L}]$; with undominated offers, $L = [Q]^+$.

(c) China offers first, the United States second: the subgame-perfect set is $L \in [L_0, \bar{L}]$; with undominated Chinese offers, $L \in [L_0, [Q]^+]$.

(d) Europe makes a take-it-or-leave-it demand of the United States, and China moves after a rejection: $L = V \cdot \mathbb{1}\{Q > 0\}$.

(e) Europe and the United States alternate offers, the status quo continues during bargaining, and Europe may exit after any rejection on the terms of China's best credible offer: in the stationary equilibrium $L \to [Q]^+$ as the discount factor approaches one.

*(f) Bargaining breaks down into exit with exogenous probability p per round (**?**): as $p \to 0$, $L \to [Q + \beta(V - Q)]^+$, where β is Europe's bargaining weight.*

Corollary 1 (A necessary condition)**.** *Under protocols (a)–(e), with undominated offers in (b) and (c), $Q \leq 0$ implies $L = 0$. The patron's dependence V raises Europe's leverage only through Europe's proposal power (d), through exogenous breakdown risk (f), or through hold-up, which we analyze in section 3.2.*

The robust content of the model, then, is the threshold. Whatever the bargaining protocol, a chokepoint holder with no credible alternative gets nothing for its chokepoint. How much of the patron's dependence it captures above the threshold depends on agenda power. Case (d) is the version of the argument that commentators have in mind when they call ASML Europe's last card: if Europe could commit to a demand, it could charge the United States the full value of the chokepoint. But a take-it-or-leave-it demand is itself a commitment, and the only threat that makes it bite is exit. Case (f) is Schelling's threat that leaves something to chance (**?**). If a dispute over ASML could slip out of control and end in realignment whether or not Europe's government wants it (say, through a change of government in The Hague), the United States must pay to avoid that risk even when $Q < 0$.

1.5 What the exit premium depends on

Proposition 3 links L to the three features of the European position this paper is about: defence dependence, the credibility of commitments, and economic dependence on China.

Proposition 3 (Determinants of leverage)**.** *Under the conditions of Proposition 1:*

(a) Defence dependence works only through conditionality. *L depends on S only through κS. If exit does not change the US stake in Europe's defence ($s_A = s_X$), then $\kappa = 0$ and no level of defence dependence gives the United States leverage over the chokepoint. Holding π_X fixed, L is nonincreasing in the reliability of the guarantee to a loyal Europe, π_A.*

(b) Compute scarcity creates and then destroys the AI lever. *As ν rises, Q falls by α when $\nu + m_{ai}$ crosses k_X, and rises by α when ν crosses k_A. Leverage is therefore non-monotone in the scarcity of compute.*

(c) Patron dependence protects the holder only against self-harming sanctions. *L is nondecreasing in V. It changes only when φV crosses $\gamma + m_f$ and firm-level penalties lose credibility, and then rises from $[Q]^+$ to $[Q + f]^+$. Jurisdiction over China-bound flows (θ) remains credible for every V.*

(d) Chinese coercion is a tax on the alliance. *The US transfer $[Q]^+$ rises one for one with $\chi_r r$ when $Q > 0$. Europe's payoff, $\max\{\omega - c, -\chi_r r\}$, is nonincreasing in r and nondecreasing in China's credible offer ω, so China's promises raise Europe's payoff only when $Q > 0$, and China's threats never do.*

(e) Unreliable US promises cause inefficient exit. *If $B_U < Q \leq V$, Europe exits even though the United States values Atlantic alignment more than China values exit.*

Part (a) turns on a distinction that alliance theory has long drawn between how much an ally depends on its patron and how much the patron can make that dependence contingent (**???**). Europe's need for US protection is enormous, but it becomes American leverage over ASML only if Washington would actually protect a loyal Europe more readily than a realigned one. The United States gets the most leverage from a guarantee that is reliable for loyal allies and void for disloyal ones, the arrangement **?** calls hierarchy. Two other kinds of guarantee give it none. An unconditional guarantee, one the United States would honour because it values European security for its own reasons, removes security from the bargain. So does a guarantee Europe no longer believes (π_A near π_X). The more doubt Washington casts on its commitment to allies who comply, the less leverage that commitment buys it over allies deciding whether to comply.

Part (b) captures a two-sided effect of the compute shortage the scenario describes. Scarcity makes denial cheap, which gives Washington a credible threat against an exiting Europe. Extreme scarcity leads Washington to ration loyal allies as well, which removes the threat's conditional part. The Frontier Inference Services Rule in the scenario, which halves allocations to a Europe that has done nothing, lowers the cost of exit at the same time as it lowers Europe's welfare.

Part (c) states precisely where the conventional view is right. The scenario observes that FDPR penalties on ASML "would also harm the US". The model agrees that this matters, but only for firm-level penalties, and only if the United States has no cheaper instrument aimed at China-bound flows. As **??** documents, it does.

Part (d) separates the two channels through which Europe's economic dependence on China operates. China's capacity to punish Atlantic alignment raises what Washington must pay, but the payment only compensates Europe for the punishment it will absorb. China's capacity to reward exit raises Europe's payoff, but only when Europe already has leverage. De-risking that reduces China's coercive reach therefore never lowers Europe's payoff. De-risking that reduces the value of China's promises lowers it only in the region $Q > 0$, which, we argue in **??** , Europe has not been in since 2019.

2 Control Rights and the Commitment Problem

The static model takes Europe's ownership of the chokepoint as given. The scenario's 2031 crisis is about ownership itself. Washington proposes to fold ASML into a joint holding company in which it holds a controlling vote over output, customers, and technology transfer. This section asks when a patron prefers to buy or coerce control of a chokepoint rather than pay rents to its holder, and what that does to the holder.

The answer turns on a commitment problem of the kind **?** and **?** identify in shifting-power settings, applied to an object whose ownership shapes future bargaining power (**?**). Europe's exit premium can change over time. Rearmament lowers κS; Chinese credit lines and market offers raise ω; years of being rationed under a tiered AI regime erode the conditional part of the AI lever (Proposition 3b). If Europe's premium is expected to turn positive, Washington faces future rents, and Europe cannot credibly promise to forgo them. Control rights remove the problem. Ownership determines who gets to choose alignment later, and the patron can buy it while its own threats still bite.

2.1 Setup

There are two periods, $t \in \{1, 2\}$, and a discount factor $\delta \in (0, 1]$. In each period in which Europe holds control, the static game of section 1 is played with period-specific parameters, giving exit premium Q_t and AI exposure α_t. We assume the AI lever is conditional in both periods ($\chi_{ai,t} = 1$) and that the United States wins each static contest ($V_t \geq Q_t + \alpha_t$, $B_U \geq V_t$).

At the start of period 1 the United States may issue an ultimatum: Europe transfers control over ASML's customers, output, and technology for both periods in exchange for a payment $T \geq 0$ made at signing; if Europe refuses, its access to frontier AI is cut in period 1. An ultimatum can be public, in which case failing to carry out the threat costs the United States an audience cost $m \in [0, \bar{m}]$ (**?**). If Europe accepts, it bears a sovereignty cost $\sigma \geq 0$, which includes the domestic political cost of surrendering a national champion and any capitalized loss of security from relocating capacity to US soil, and China punishes the transfer with $r^c \geq 0$. Once control has passed, the United States has no reason to cut Europe off: its ex post gain from doing so is negative, so the assurance half of the ultimatum is credible (**?**). If Europe refuses, the United States decides whether to cut. Cutting a Europe that remains aligned yields the direct ex post gain $g_N = \nu_1 - k_A$, which is negative outside the extreme-scarcity regime. It also removes the AI term from Europe's period-1 exit costs, so the period-1 exit premium rises from Q_1 to $Q_1 + \alpha_1$.

Payoffs, normalized as before, are:

Outcome	Europe	United States
No ultimatum	$[Q_1]^+ + \delta[Q_2]^+$	$-[Q_1]^+ - \delta[Q_2]^+$
Accept	$T - \sigma - r^c$	$-T$
Refuse, cut	$[Q_1 + \alpha_1]^+ - \alpha_1 + \delta[Q_2]^+$	$g_N - [Q_1 + \alpha_1]^+ - \delta[Q_2]^+$
Refuse, no cut	$[Q_1]^+ + \delta[Q_2]^+$	$-m - [Q_1]^+ - \delta[Q_2]^+$

Access is restored in period 2; Lemma 3 shows why.

2.2 The bite of a credible threat

Lemma 2 (Credible ultimatum and effective bite)**.** *After a refusal, the United States cuts Europe off if and only if*

$$m + g_N \;\geq\; [Q_1 + \alpha_1]^+ - [Q_1]^+. \tag{8}$$

When it does, Europe's net loss is the effective bite

$$b_1 \equiv \alpha_1 - \left([Q_1 + \alpha_1]^+ - [Q_1]^+\right) = \min\{\alpha_1, [-Q_1]^+\}. \tag{9}$$

Two features matter. First, without an audience cost the threat is not credible: with $m = 0$ and $g_N < 0$, cutting off a Europe that has merely refused to sell control costs Washington something and gains it nothing. That is why the scenario's White House has to go public. Second, a punishment hurts its target only up to the target's slack. Cutting AI access removes one of the reasons Europe stays aligned. Once $Q_1 + \alpha_1 > 0$, the rest of the cut comes back to Europe as rent, because Washington must now pay to keep it from exiting. A large stick adds nothing beyond the point where using it would make exit attractive, which is the logic of marginal deterrence (**?**).

Lemma 3 (Self-limiting sanctions)**.** *Extending the cut into period 2 costs the United States* $[Q_2 + \alpha_2]^+ - [Q_2]^+$ *in additional rent, which equals* α_2 *when* $Q_2 \geq 0$*. A permanent cut is credible only if the period-2 audience cost exceeds this amount less* g_N*.*

The scenario's threat that Europe will "lose all access, present and future, to American AI" is therefore not credible inside the model unless Washington has staked a great deal of reputation on it. A permanent cut hands Europe exactly the exit option the cut was meant to deter, at the moment Europe's future premium is positive.

2.3 Preventive control

Proposition 4 (Preventive control)**.** *Under complete information, the United States issues a public ultimatum, and Europe accepts, if and only if*

(i) Europe's future exit premium is positive, $Q_2 > 0$*;*

(ii) the threat can be made credible, $\bar{m} + g_N \geq [Q_1 + \alpha_1]^+ - [Q_1]^+$*; and*

(iii) the effective bite exceeds the deadweight cost of transferring control, $b_1 > \sigma + r^c$*.*

The payment is $T^* = [\delta Q_2 + \sigma + r^c - b_1]^+$ *and Europe's payoff is* $\max\{\delta Q_2 - b_1,\ -\sigma - r^c\}$*.*

Because $b_1 \leq [-Q_1]^+$, condition (iii) requires $Q_1 < -(\sigma + r^c)$. Washington must act while Europe's exit premium is still well below zero, and it acts because that premium is expected to become positive. A patron that faces no future rents has no reason to seek control; a patron whose threats no longer bite cannot get it. Preventive control occupies the interval between those two states. The same logic extends to many periods.

Corollary 2 (Windows)**.** *In a* T*-period version in which the United States may issue an ultimatum at any date at which Europe still holds control, it does so at some date if and only if there is a date* t *at which the cut is credible and* $b_t > \sigma + r^c$*, and a later date* $s > t$ *with* $Q_s > 0$*.*

The US window, the set of dates at which (ii) and (iii) hold, and Europe's window, the dates with $Q_s > 0$, are separated by at least $\sigma + r^c$ in units of the exit premium. Under complete information the speed of the shift does not matter, only its direction and the existence of a window of credible coercion before Europe's window opens. Two features of Proposition 4 cut against intuition. China's threat to punish the transfer, r^c, protects European ownership, because it raises the bite Washington needs. And Europe's sovereignty costs, which commentators treat as an obstacle to a deal, play the same role: a Netherlands that would pay a high domestic price for surrendering ASML is harder to coerce into doing so.

2.4 Private sovereignty costs and breakdown

The scenario's endgame is a standoff, not a signature. Under complete information the ultimatum is either accepted or never issued. Breakdown requires that Washington be uncertain about what surrender would cost the European governments involved, which is plausible given coalition politics in The Hague and Berlin. Let σ be Europe's private information, uniformly distributed on $[0, \bar{\sigma}]$, and suppose conditions (i) and (ii) of Proposition 4 hold. Write $R \equiv \delta Q_2$, $a \equiv b_1 - r^c > 0$, and $e \equiv -g_N + [Q_1 + \alpha_1]^+ \geq 0$ for what it costs Washington to carry out the cut.

Proposition 5 (Breakdown)**.** *If the United States issues the ultimatum, Europe accepts if and only if $\sigma \leq s^*$, where*

$$s^* = \max\Big\{a - R,\ \min\big\{\bar{\sigma},\ \tfrac{a+e}{2}\big\}\Big\}. \tag{10}$$

With probability $P_b = [1 - s^*/\bar{\sigma}]^+$ Europe refuses, the cut is carried out, and Europe keeps ASML. P_b is nonincreasing in the bite b_1 and in the execution cost e, and nondecreasing in China's punishment of the transfer r^c and in $\bar{\sigma}$. When $a - R > \min\{\bar{\sigma}, (a+e)/2\}$, the United States offers nothing and P_b is increasing in Europe's future premium R.

Breakdown is the scenario's "neither" outcome: Europe refuses both great powers, loses frontier access, and keeps the chokepoint. The proposition says when it becomes likely. It is more likely when China credibly threatens to punish a transfer, when the European domestic cost of surrender is uncertain, and when Washington's threat is cheap to execute. The last point is the usual risk-return trade-off (**??**): a patron that loses little from a standoff makes tougher offers and is refused more often.

2.5 Promises or threats

Europe could try to remove the commitment problem itself. An allied export-control compact, dual-key licensing that gives Washington a veto over China-bound sales, or treaty-entrenched EU controls would bind Europe to Atlantic alignment in period 2 without transferring ownership. Suppose Europe can adopt such a compact unilaterally before period 1, at cost σ_c, and China punishes it with r^b.

Remark 1 (Self-binding)*.* When the conditions of Proposition 4 hold, Europe strictly prefers to bind itself if and only if $\sigma_c + r^b < \sigma + r^c$ and $\sigma_c + r^b < b_1 - \delta Q_2$. Binding removes the US motive for an ultimatum and keeps ownership in Europe, but it forfeits the future rents that a positive Q_2 would have earned.

A credible promise not to exit and a credible exit threat cannot coexist: the compact works by setting Europe's future exit premium to minus infinity. It does not remove the other source of leverage, the threat to withhold servicing from the United States itself, which section 3.2 analyzes. Binding against exit also removes the preventive part of Washington's stake, which can make the United States more willing to concede a hold-up dispute.

3 Extensions

The baseline treats Europe as one actor, gives it a single defection option, and lets the United States keep its chips flowing while the two sides bargain. Each extension relaxes one of these assumptions. The first asks who in Europe actually decides. The second gives Europe the power to hurt the United States during a dispute, which is where the patron's dependence on ASML returns. The third adds partial defection.

3.1 Who decides: nested chokepoints and veto players

Three features of the real decision structure matter. The Netherlands licenses ASML's exports to China, but a licence permits a shipment; it does not order one. ASML itself decides whether to ship, and its own exposure to US law, customers, suppliers, and shareholders is large. Turning ASML against the United States, finally, is not a Dutch decision at all: exports to the United States move under an EU general authorization, and the instrument designed for retaliation, the Anti-Coercion Instrument (ACI), requires qualified-majority decisions in the Council.

Let the firm's China profit from exit be y and its expected loss from US measures aimed at it be f^F. If $y < f^F$, the firm will not ship unless the state compels it at cost $k_s \geq 0$, for instance under emergency powers like those the Dutch government used against Nexperia in 2025. For the veto-player results, let each member state i have its own exit premium $Q_i = \omega_i - c_i + \chi_{r,i} r_i$, computed from its own share of China's offer and its own exposure to US and Chinese instruments. For the ACI, let q_i be member i's net gain from adopting a countermeasure against the United States, and let $\mathcal{B}$ be the family of blocking coalitions: at least four members holding more than 35 percent of EU population.

Proposition 6 (Nested chokepoints)**.** *(a)* Firm compliance. *Europe's effective exit premium is $Q - k_s \mathbb{1}\{y < f^F\}$. If the United States can raise f^F above y at negligible cost, Europe's leverage is at most $[Q - k_s]^+$.*

(b) Holder veto. *If exit requires the consent of member h alone, the United States pays $[Q_h]^+$. A conditional benefit granted to h alone and withdrawn upon exit raises c_h one for one and lowers this payment.*

(c) Blocking minorities. *If no one bids for adoption, the United States defeats an ACI countermeasure at cost $K(0) = \min_{S \in \mathcal{B}} \sum_{i \in S} [q_i]^+$. If a rival bidder values adoption at $B > 0$ and moves second, the United States must make every qualified majority cost at least B; its cost $K(B) = \min_{\tau \geq 0} \sum_i \tau_i$ subject to $\sum_{i \in S} [\tau_i - q_i]^+ \geq B$ for every qualified majority S can exceed $K(0)$ by more than B and generally requires payments to members outside any minimal blocking coalition (**?**).*

Part (a) is the institutional point in **?** turned around. A state can weaponize a network hub only if its domestic institutions give it control over the hub. ASML meets the network condition but sits inside networks the United States controls: US-origin subsystems, US persons, and US customers. Europe's chokepoint is nested in American ones, and the exit threat is only as credible as the Dutch state's willingness to compel its own champion to break US law. Part (b) is divide and rule. Granting the Netherlands, and only the Netherlands, preferential AI access, as the scenario's tier system does (the real January 2025 AI Diffusion Rule similarly split the EU, placing ten members in its top tier for reasons unrelated to ASML), gives the veto player more to lose from exit than its neighbours. Part (c) corrects a piece of arithmetic. Under the 2026 Council population figures (**?**), the states the scenario says opposed or abstained on the 2029 countermeasure (the Netherlands, Ireland, Poland, the three Baltic states, and Italy) hold 28.05 percent of EU population, short of a blocking minority, and any blocking coalition that excludes Germany, France, Italy, and Spain needs at least twelve members. The pivotal members on a measure against the United States are large states, not the eastern flank.

3.2 Voice through hold-up: time to pain

The baseline assumes that bargaining takes place under the status quo, with ASML servicing US-aligned fabs throughout. The scenario's 2031 standoff is different. Once Europe refuses the

ultimatum, Washington cuts off frontier AI, and Europe could interrupt servicing, spare parts, and tool deliveries in return. The scenario claims that Washington "has amassed enough EUV machines and maintenance know-how to forego ASML access for longer than Europe can forego access to frontier AI". That is a claim about time to pain, and the patron's dependence on the chokepoint enters through it.

Let Europe and the United States dispute a divisible issue: the scope of US control over ASML, or the terms of Europe's AI access. An agreement leaves Europe a share $p \in [0,1]$, with payoffs $-(1-p)s_E$ to Europe and $(1-p)s_U$ to the United States. Until they agree, Europe bears a flow cost c_E from date τ_E on, and the United States bears c_U from date τ_U on. The buffers are what each side can live on before the cutoff hurts. For the United States, that means stockpiled tools, spare parts, and transferred maintenance know-how. For Europe, it means compute on European soil and stored model weights. Offers arrive at intervals Δ; the United States proposes with probability ψ and Europe otherwise. We take $\Delta \to 0$ and assume Europe's exit option is unattractive. Write $\hat{c}_i \equiv c_i/s_i$ and $D \equiv \tau_U - \tau_E$.

Proposition 7 (Time to pain)**.** *Suppose $\psi\hat{c}_E \neq (1-\psi)\hat{c}_U$.*

(a) The equilibrium is unique and agreement is immediate. If Europe is the less vulnerable side once both buffers are exhausted, $\psi\hat{c}_E < (1-\psi)\hat{c}_U$, its share is $p^* = 1$ when $D \leq 0$ and $p^* = [1-\psi\hat{c}_E D]^+$ when $D > 0$. If instead $\psi\hat{c}_E > (1-\psi)\hat{c}_U$, then $p^* = 0$ when $D \geq 0$ and $p^* = \min\{1, (1-\psi)\hat{c}_U|D|\}$ when $D < 0$.

(b) Let $\tau_E = 0$ and suppose the United States bears no cost unless Europe interrupts servicing. Interruption, which raises Europe's flow cost to $c_E + \ell$ and imposes c_U on the United States after τ_U, is played in a Markov perfect equilibrium if and only if $(1-\psi)\hat{c}_U > \psi(c_E+\ell)/s_E$ and $\tau_U < H_E \equiv s_E/[\psi(c_E+\ell)]$. Europe's share is then $1 - \tau_U/H_E$.

The proposition separates the two quantities **?** called sensitivity and vulnerability. The buffer gap D measures sensitivity: who feels the cutoff first. The post-buffer ranking of normalized costs measures vulnerability: who could hold out longer once neither has a cushion. Hold-up gives Europe leverage only if it wins on vulnerability and the US buffer is shorter than Europe's endurance horizon H_E. In the interior, each unit of US buffer costs Europe ψc_E, Europe's own rate of pain weighted by US agenda power; the US cost c_U matters only for who wins on vulnerability. This is where V re-enters the analysis. The more US frontier compute depends on continued ASML servicing, the larger c_U. The scenario's claim that a stockpile lets Washington outlast Europe amounts to the claim that $\tau_U \geq H_E$. Its Epilogue's call for compute on European soil amounts to raising τ_E. Servicing and spare parts matter more than tools here: installed EUV machines keep running, but they need engineers and consumable parts, so the US buffer for servicing is short even if its stock of tools is large.

3.3 Hedging and marginal deterrence

The scenario's 2028 crisis, like the real disputes of 2023–2025, was not about exit. It was about whether ASML could keep selling and servicing older immersion tools in China. Add a third alignment, hedging (H), with China's credible offer $\omega_H < \omega$ and exit costs c_H, and suppose the United States, after observing a hedge, may withhold an instrument k that also deters exit. Let D_k be the instrument's contribution to Europe's exit costs (κS for security, α for AI) and $g_k(H)$ the United States' ex post gain from withholding it from a hedging Europe.

Proposition 8 (Marginal deterrence)**.** *Withholding instrument k to punish hedging is credible if and only if*

$$m_k + g_k(H) \;\geq\; [Q + D_k]^+ - [Q]^+. \tag{11}$$

Enforcing jurisdiction over the hedged flows themselves leaves Q unchanged and is credible whenever its direct cost is small. If $Q + D_k > 0$ and audience costs are small, the United States cannot use security or AI linkage against hedging, and must rely on flow-specific jurisdiction.

A patron cannot spend the sanctions that deter full defection on partial defection without weakening deterrence of the full version (**??**). The prediction is that Washington polices legacy-tool sales with targeted jurisdiction over those sales (de minimis rules for lithography tools bound for China, the foreign direct product rule, servicing restrictions) and keeps security and AI linkage in reserve. This matches the scenario's 2028 crisis, where the threat is the FDPR, and it matches the actual US sequence from 2019 to 2024. Jurisdictional threats of this kind also work under uncertainty about US resolve. If the firm-level penalty f_H is existential for ASML while the revenue at stake is a fraction of its sales, even a small probability that Washington means it is enough. That is the formal content of the scenario's line that ASML "cannot afford to call Washington's bluff".

A Proofs

Throughout, $[x]^+ = \max\{0, x\}$. Payoffs are normalized as in section 1.1.

Proof of Lemma 1

Stage (4) decisions are taken after Europe's alignment is fixed and after transfers have been paid, so each decision maximizes the decision maker's ex post payoff. (i) In a crisis with cost K, defending yields the United States $s_a - K$ and not defending yields zero, so it defends iff $s_a \geq K$, an event of probability $G(s_a) = \pi_a$. (ii) After exit, denial yields $\nu - k_X$ plus avoidance of the audience cost m_{ai} attached to the public denial rule; supplying yields zero. Denial occurs iff $\nu + m_{ai} \geq k_X$. After Atlantic alignment no commitment is at stake and denial occurs iff $\nu \geq k_A$. Because $k_A > k_X$ and $m_{ai} \geq 0$, denial after A implies denial after X, so the conditional part of the instrument is $\chi_{ai} = \mathbb{1}\{\nu + m_{ai} \geq k_X\}\mathbb{1}\{\nu < k_A\}$. (iii) After exit, penalizing ASML yields $\gamma - \varphi V$ and failing to do so costs m_f; the United States penalizes iff $\varphi V \leq \gamma + m_f$. After Atlantic alignment, penalizing yields $-\varphi V < 0$ and no commitment is at stake. (iv) After A, punishment yields China g_r and avoids audience cost m_r; it punishes iff $g_r + m_r \geq 0$. After X China has obtained what it sought and punishment has no rationale. Europe's expected security loss after a is $(1 - \pi_a)S$, so exit adds $(\pi_A - \pi_X)S = \kappa S$. The other terms follow directly, and terms common to both alignments (for instance, AI denial when $\nu \geq k_A$) cancel. □

Proof of Proposition 1

Fix the grid mesh $\varepsilon > 0$ and assume B_U and B_C are grid points. By Lemma 1 and eq. (5), Europe chooses A iff $\tau_U - \chi_r r \geq \lambda_C \tau_C - c$.

China's best response. Given τ_U, China wins only with an offer τ_C satisfying $\lambda_C \tau_C > \tau_U - \chi_r r + c$. Let $\underline{\tau}_C(\tau_U)$ be the smallest grid point with this property. Winning yields $(1 - \theta)W - \underline{\tau}_C(\tau_U)$, losing yields zero, and China does not bid at zero profit. China therefore wins iff $\underline{\tau}_C(\tau_U) \leq B_C$ and $\underline{\tau}_C(\tau_U) < (1 - \theta)W$. Since $\underline{\tau}_C(\tau_U) \in \big((\tau_U - \chi_r r + c)/\lambda_C,\ (\tau_U - \chi_r r + c)/\lambda_C + \varepsilon\big]$, China wins whenever $\tau_U < Q - \lambda_C \varepsilon$ and loses whenever $\tau_U \geq Q$.

The US offer. Any $\tau_U \geq [Q]^+$ that is a grid point keeps Europe aligned. The United States' cheapest such offer, τ_U^ε, satisfies $[Q]^+ - \lambda_C \varepsilon \leq \tau_U^\varepsilon \leq [Q]^+ + \varepsilon$ (the lower bound because offers in $[Q - \lambda_C \varepsilon, Q)$ may also deter China, depending on where the grid falls). Keeping Europe aligned at this price is better for the United States than losing V iff $\tau_U^\varepsilon \leq V$, and feasible iff $\tau_U^\varepsilon \leq B_U$. Under $Q \leq \min\{V, B_U\}$ this holds in the limit, so every subgame-perfect equilibrium has $\tau_U^\varepsilon \to [Q]^+$ and Europe choosing A. Payoffs follow from eq. (5). If $Q > \min\{V, B_U\}$, for small ε the United States either cannot or will not offer Q, and China wins. □

Proof of Proposition 2

Write $\bar{L} \equiv \min\{V, [\lambda_C B_C - c + \chi_r r]^+\}$, the largest US transfer a Chinese offer within the cap can force. Under the stated assumptions, $Q \leq \bar{L} \leq V$.

(a) This is Proposition 1.

(b) In any Nash equilibrium in which Europe chooses A, the US offer must satisfy $\tau_U \geq [Q]^+$; otherwise China could deviate to a profitable winning offer. The US offer must also equal $[\lambda_C \tau_C - c + \chi_r r]^+$ for China's actual offer τ_C; otherwise the United States could lower its offer and still win. Any $\tau_C \leq B_C$ with $\lambda_C \tau_C - c + \chi_r r \in [[Q]^+, \bar{L}]$ supports such an equilibrium, with the United States offering exactly that amount: China loses, earns zero, and cannot profitably win because $\tau_U \geq [Q]^+$; the United States prefers matching to losing because the required transfer is at most V. China's offers above $(1-\theta)W$ are weakly dominated, because if accepted they yield a loss. With undominated offers, $\lambda_C \tau_C - c + \chi_r r \leq Q$, so $L = [Q]^+$.

(c) Given τ_C, the United States offers $[\lambda_C \tau_C - c + \chi_r r]^+$ if that is at most V, and otherwise lets Europe exit. Because $Q \leq V$, China cannot win at a price below its valuation, so China earns zero and is indifferent among all losing offers. Every $\tau_C \in [0, B_C]$ with $\lambda_C \tau_C - c + \chi_r r \leq V$ is part of a subgame-perfect equilibrium, which gives $L \in [L_0, \bar{L}]$, with $L_0 = [\chi_r r - c]^+$ obtained at $\tau_C = 0$. Excluding weakly dominated offers above $(1-\theta)W$ gives $L \in [L_0, [Q]^+]$.

(d) After a US rejection, China can induce exit at a profit iff $Q > 0$, by offering slightly more than $(c - \chi_r r)/\lambda_C$. If $Q > 0$ the United States accepts any demand up to V, so Europe demands V. If $Q \leq 0$ a rejection leaves Europe aligned with no transfer, so the United States rejects every positive demand.

(e) The status quo gives the United States the alignment it wants and gives Europe nothing, so absent exit the United States accepts no positive transfer and the inside-option payoff is zero. Exit yields Europe $\omega - c$ per period against $-\chi_r r$ under the status quo. Suppose $Q > 0$. In the stationary equilibrium the United States, when it proposes, offers the per-period transfer Q, the least that stops Europe from rejecting and exiting. When Europe proposes, the United States accepts a transfer τ forever iff $-\tau/(1-\delta) \geq -\delta Q/(1-\delta)$, so Europe demands δQ. Both converge to Q as $\delta \to 1$. If $Q \leq 0$, exit is not a credible response to a rejection and the United States accepts no positive transfer. This is the outside-option principle (**??**).

(f) With exogenous breakdown into exit, the limit of the alternating-offers game as $p \to 0$ is the Nash bargaining solution with breakdown payoffs as the disagreement point (**?**): $-V$ for the United States and $\omega - c$ for Europe. Maximizing $(V - \tau)^{1-\beta}(\tau - \chi_r r - \omega + c)^\beta$ gives $\tau = Q + \beta(V - Q)$, truncated at zero. □

Proof of Proposition 3

(a) By eq. (6), S enters only through κS, and $\kappa = G(s_A) - G(s_X) = 0$ when $s_A = s_X$. Holding π_X fixed, $\partial Q/\partial \pi_A = -S \leq 0$, and $[Q]^+$ is nondecreasing in Q. (b) χ_{ai} switches from 0 to 1 at $\nu = k_X - m_{ai}$ and from 1 to 0 at $\nu = k_A$, with $k_X - m_{ai} < k_A$. (c) V enters Q only through

$\chi_f = \mathbb{1}\{\varphi V \le \gamma + m_f\}$, which is nonincreasing in V; when it switches from 1 to 0, Q rises by f. (d) By Proposition 1 Europe's payoff is $[Q]^+ - \chi_r r = \max\{-\chi_r r, \omega - c\}$. (e) See the last step of the proof of Proposition 1. □

Proof of Lemmas 2 and 3

After a refusal, cutting yields the United States $g_N - [Q_1 + \alpha_1]^+ - \delta[Q_2]^+$ and not cutting yields $-m - [Q_1]^+ - \delta[Q_2]^+$; comparing gives eq. (8). Europe's payoff falls by $\alpha_1 - ([Q_1 + \alpha_1]^+ - [Q_1]^+)$. If $Q_1 \ge 0$ this is zero; if $-\alpha_1 < Q_1 < 0$ it is $-Q_1$; if $Q_1 \le -\alpha_1$ it is α_1. Hence $b_1 = \min\{\alpha_1, [-Q_1]^+\}$. The period-2 statement follows by the same comparison applied to period 2. □

Proof of Proposition 4

Suppose (ii) holds, so the United States can choose m large enough to make the cut credible. Europe's refusal payoff is $\rho_E = [Q_1]^+ - b_1 + \delta[Q_2]^+$, and it accepts iff $T - \sigma - r^c \ge \rho_E$. The cheapest acceptable payment is $T^* = [\rho_E + \sigma + r^c]^+$. The United States issues the ultimatum iff $-T^* > -[Q_1]^+ - \delta[Q_2]^+$, i.e., iff its gain $\Gamma \equiv [Q_1]^+ + \delta[Q_2]^+ - T^*$ is positive.

If $b_1 = 0$, then $T^* \ge [Q_1]^+ + \delta[Q_2]^+$ and $\Gamma \le 0$. If $b_1 > 0$, then $Q_1 < 0$, so $\Gamma = \delta[Q_2]^+ - [\delta[Q_2]^+ - b_1 + \sigma + r^c]^+ = \min\{\delta[Q_2]^+, b_1 - \sigma - r^c\}$, which is positive iff $Q_2 > 0$ and $b_1 > \sigma + r^c$. If (ii) fails, refusal costs Europe nothing, $T^* = [Q_1]^+ + \delta[Q_2]^+ + \sigma + r^c$, and $\Gamma \le 0$. Europe's payoff when the ultimatum is issued is $T^* - \sigma - r^c = \max\{\delta Q_2 - b_1, -\sigma - r^c\}$. □

Proof of Corollary 2

Call a date t *eligible* if the cut is credible at t, $b_t > \sigma + r^c$, and $Q_s > 0$ for some $s > t$. In any subgame starting at a date with no later eligible date, the argument in the proof of Proposition 4, with $\delta[Q_2]^+$ replaced by the discounted sum of later rents, shows that no ultimatum is issued after that date. Let t^* be the last eligible date. In the subgame beginning at t^* with Europe in control, the continuation after a refusal contains strictly positive rents and no further ultimatum, so the United States' gain from issuing is $\min\{\text{discounted future rents}, b_{t^*} - \sigma - r^c\} > 0$, and it issues. On the equilibrium path an ultimatum is therefore issued at t^* or earlier. Conversely, if no date is eligible, then by backward induction no subgame contains an ultimatum: at each date either the cut is not credible, or the bite is too small, or there are no later rents to remove, and in each case the gain from issuing is nonpositive. We also verified the statement numerically on 100,000 random parameter draws with up to six periods (appendix B). □

Proof of Proposition 5

With conditions (i)–(ii) of Proposition 4, the United States sets $m \ge e$, so it carries out the cut after a refusal. Europe accepts iff $T - \sigma - r^c \ge -b_1 + R$, i.e., iff $\sigma \le s \equiv T + a - R$. Relative to issuing no ultimatum, the United States gains $R - T$ if Europe accepts and loses e if Europe refuses. Write $x = R - T \le R$, so $s = a - x$. For $s \in (0, \bar\sigma)$ its expected gain is

$$\Pi(x) = \frac{(a - x)\,x - (\bar\sigma - a + x)\,e}{\bar\sigma},$$

which is strictly concave with unconstrained maximizer $x^\circ = (a - e)/2$, so $s^\circ = (a + e)/2$. If $s \ge \bar\sigma$, every type accepts and $\Pi = x$, which is increasing in x, so the United States never sets $s > \bar\sigma$. Imposing $s \le \bar\sigma$ and $x \le R$ gives $s^* = \max\{a - R, \min\{\bar\sigma, (a + e)/2\}\}$. The comparative statics follow because s^* is nondecreasing in $a = b_1 - r^c$ and in e, and $P_b = [1 - s^*/\bar\sigma]^+$. In the corner

$s^* = a - R$, s^* is decreasing in R. The United States issues the ultimatum iff $\Pi(x^*) > 0$; in the interior this is $(a+e)^2 > 4\bar{\sigma}e$, and with $e = 0$ it reduces to $a > 0$. □

Proof of Remark 1

If the conditions of Proposition 4 hold, Europe's payoff without binding is $\max\{\delta Q_2 - b_1, -\sigma - r^c\}$. With binding, Europe has no future exit premium, the United States gains nothing from an ultimatum, and Europe's payoff is $-\sigma_c - r^b$ (period-1 rents are zero because $Q_1 < 0$). Binding is strictly better iff $-\sigma_c - r^b$ exceeds both terms. □

Proof of Proposition 6

(a) If $y < f^F$ the firm refuses to ship after a licence, and exit occurs only if the state compels it, which costs k_s; substituting into Proposition 1 gives the result. (b) China needs only h's consent, so the argument of Proposition 1 applies to h's payoffs. A benefit Δ that h receives under A and loses under exit raises c_h by Δ. (c) With no rival bidder, a countermeasure is adopted iff the members with $q_i > \tau_i$ form a qualified majority, which the United States prevents most cheaply by paying $[q_i]^+$ to every member of some blocking coalition. With a rival bidder moving second, the rival buys a qualified majority S at cost $\sum_{i \in S}[\tau_i - q_i]^+$ whenever that is less than B, which gives the linear program. Example: three members, adoption by two, $q_i = 0$. Then $K(0) = 0$, while $K(B)$ requires $\tau_i + \tau_j \geq B$ for every pair, so $K(B) = 3B/2$ with $\tau_i = B/2$ for all three. □

Proof of Proposition 7

(a) Let $w(t)$ be Europe's expected share at date t before the proposer is drawn. A proposer offers the responder exactly its continuation value, and agreement is immediate because delay is weakly costly to both. Hence

$$w(t) = \psi \max\{w(t+\Delta) - \hat{c}_E(t)\Delta,\, 0\} + (1-\psi)\min\{1,\, w(t+\Delta) + \hat{c}_U(t)\Delta\}.$$

After both buffers are exhausted the costs are constant; in the interior $w(t) - w(t+\Delta) = [(1-\psi)\hat{c}_U - \psi\hat{c}_E]\Delta$, so in an infinite horizon the only bounded solution is $w = 1$ if $(1-\psi)\hat{c}_U > \psi\hat{c}_E$ and $w = 0$ otherwise (cf. the fixed-cost case of **?**). Between the two buffer dates only one side bears costs. If $D > 0$, Europe alone bears costs on $[\tau_E, \tau_U)$ and w falls (moving forward in time) at rate $\psi\hat{c}_E$, truncated at zero; if $D < 0$, the United States alone bears costs on $[\tau_U, \tau_E)$ and w rises at rate $(1-\psi)\hat{c}_U$, truncated at one. Before the earlier buffer date nobody bears costs and w is constant. Taking $\Delta \to 0$ gives the stated shares. (b) Without interruption the United States bears no cost and Europe bears c_E from date zero, so $p^* = 0$. With interruption, part (a) with $\tau_E = 0$ gives $p^* = [1 - \psi\tau_U(c_E + \ell)/s_E]^+ = [1 - \tau_U/H_E]^+$ if Europe is the less vulnerable side post-buffer, and $p^* = 0$ otherwise. Interruption is played iff it raises p^*. □

Proof of Proposition 8

After a hedge, the United States compares withholding k, which yields $g_k(H)$ but removes k from Europe's exit costs, so that the subsequent static contest costs it $[Q + D_k]^+$, with not withholding, which costs m_k if it had committed to punish hedging and leaves the contest at $[Q]^+$. Jurisdiction over hedged flows does not enter Europe's exit costs, so enforcing it leaves the contest at $[Q]^+$. □

B Numerical verification

We checked each result by brute force in Python, with scripts in the replication folder. Proposition 1 was checked by backward induction on a grid of offers (mesh 0.01) over random parameter draws with $V \geq Q$. Proposition 4 was checked on 200,000 random draws with ties never broken in favour of the threat, and Corollary 2 on 100,000 draws with two to six periods; both produced zero mismatches. Proposition 5 was checked against direct numerical optimization of the US offer. Proposition 7 was checked by backward induction in discrete time with $\Delta = 0.001$; deviations from the closed form are $O(\Delta)$ once the stationary phase is long enough to reach its boundary value.

References